

Infosys Springboard Virtual Internship 7.0 Completion Report

Batch Number 1 : Enterprise Learning Platform

Start date : 29-06-2026

Name: M.Madhumitha Kumari

Internship Duration: 40-day (12 Weeks)

1. Project Title

Enterprise Learning Platform

2. Project Objective

Enterprise Learning Platform is an AI-powered enterprise learning platform designed to help users monitor and improve their technical skills, career readiness, productivity, and daily learning habits.

The main objectives of this project are:

- To provide users with an interactive AI chatbot and code tutor for learning support.
- To analyze user learning data and provide insights into their skill patterns.
- To predict mood, stress levels, productivity, and career elevation paths using machine learning & AI models.
- To provide personalized learning suggestions to improve technical and career performance.
- To create a dashboard that visualizes user learning analytics in an easy-to-understand format.

This project aims to combine artificial intelligence and data analytics to promote better skill awareness and learning management.

3. Project description in detail

Enterprise Learning Platform is a web-based AI enterprise learning platform developed using React 19, Spring Boot 3.5 (Java 17), MySQL 8.0, and Google Gemini AI. The platform allows users to track their daily learning activities, analyze skill progress data, and receive intelligent recommendations. The system consists of multiple modules that work together.

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Landing Page:

The landing page introduces users to the platform and highlights the key features of Enterprise Learning Platform. It provides information about the purpose of the platform and encourages users to register or log in.

Authentication System:

The system includes a secure login and registration mechanism using JWT sessions and Google OAuth2 SSO that allows users to access their personal learning dashboard.

Learning Dashboard:

The dashboard is the main interface where users can view their learning insights. It displays charts and statistics about course progress, XP points, study streaks, and skill badges.

AI Chatbot:

The chatbot interacts with users and provides technical advice based on their inputs. It acts as a virtual assistant and code tutor that can guide users in maintaining a disciplined learning path.

Machine Learning Model & AI Career Engine:

The platform uses machine learning and AI models built using Scikit-learn and Google Gemini to predict skill growth, stress levels, and productivity.

The model is trained on a learning dataset that includes features such as:

- Course completion rate
- Quiz evaluation scores
- Coding sandbox submission accuracy
- Daily study streak duration
- Skill gap benchmark score

The predictions help users understand their technical competency and career path readiness.

Admin Panel & Instructor Studio:

An admin panel allows administrators to monitor system activity, manage users, review learning analytics, and control platform access. The entire system is designed to provide a simple, user-friendly interface while using advanced technology in the backend.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project objective is decided, making github repository	Github repository created, finalising teams with the project overview
Week 2	Tech stack evaluation, architecture planning	Spring Boot & React setup completed, learning about REST APIs
Week 3	Setup user authentication, JWT tokens	Done with verified signin login with JWT sessions

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Week 4	Learning about API's , setting api's for project , database design and SQL commands	API's is setup using openAi , gemini etc. which makes our project more effective
Week 5	LMS pipeline development – course hierarchy, content mapping	Tiered course catalog (Basecamp, Ridge, Summit) implemented
Week 6	Spring Boot service logic, repository interfaces	Implementing service layer functions and MySQL JPA mappings
Week 7	Gemini AI integration, prompt template design	Used Gemini models for live AI assistant & code tutor in Enterprise Learning Platform
Week 8	Quiz assessment, assignment portal development.	Automated quiz scoring and assignment upload portal completed.
Week 9	Gamification system - XP, badges, leaderboards.	Displayed leaderboards, XP points, and user badges based on activity.
Week 10	Dark/Light theme switch, UI polishing.	Theme toggle working across UI, modern glass morphism aesthetic.
Week 11	Dashboard design (like user activity, streak, profile, feedback).	Implementing dashboard with different course tiers and activity heatmap.
Week 12	Adding career suite (ATS Resume, digital certs, admin	Project fully tested and ready for evaluation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined the project objectives, scope, system architecture, and responsibilities for developing Enterprise Learning platform.	29 June2026
Prototype / First Draft	Developed initial working prototype including Spring Boot backend, basic UI pages, and core course input functionality.	08 Jul 2026
Mid-Term Review	Evaluated project progress, improved dashboard design, integrated Gemini AI model, and refined LMS functionality.	23 Jul 2026
Final Submission	Completed the fully functional SkillSphere system including dashboard analytics, AI chatbot, gamification, and certs.	13 Aug 2026
Final Presentation	Demonstrated the working system highlighting features such as wellness tracking, predictions, chatbot interaction, and system architecture.	02 Sep 2026

5b. Project execution details

1.Requirement Analysis:

The project started with identifying the main objectives of the Enterprise Learning Platform system, which included building an AI- powered learning assistant that can track user progress data, provide chatbot support, and generate predictions related to skills, streak, and career readiness.

2.System Architecture Design:

The architecture of the system was planned using a client-server model where the React SPA frontend interacts with the Spring Boot backend REST controllers. The backend processes user requests, communicates with the MySQL database, and integrates the Google Gemini AI engine.

3.Frontend Development:

The user interface was developed using HTML, CSS, JavaScript, and React 19. Multiple pages were created including the landing page, login page, dashboard page, course player, and chatbot interface to provide a smooth and user-friendly experience.

4. Backend Development:

The backend was implemented using Spring Boot 3.5 and Java 17. It handles user authentication, manages API routes, processes data, connects to the MySQL database, and communicates with the AI module.

5.Database Integration:

MySQL database was used to store user credentials, course records, quiz scores, and prediction data. This ensures that all user information and analytics can be accessed and managed efficiently.

6.Machine Learning & AI Integration:

Google Gemini AI and Scikit-learn models were integrated into the Spring Boot backend so that AI tutoring responses and career roadmaps can be generated in real-time when users input their learning data.

7.LMS & Interactive Playground Integration:

Structured Basecamp, Ridge, and Summit course routes were built with video streaming, markdown reading, and interactive coding exercise sandboxes.

8.Dashboard Development:

An interactive dashboard was created using Chart.js to visualize learning data through graphs and charts. This helps users easily understand their progress trends and performance.

9.Chatbot Integration:

An AI chatbot module was added to allow users to interact with the system. The chatbot provides technical advice and guidance based on user inputs and evaluation results.

10.Admin Panel and Testing:

An admin panel was implemented to monitor users and manage system activities. Finally, the entire system was tested to ensure that all modules work correctly and integrate smoothly.

6. Snapshots / Screenshots

Opening the project folder containing all the main files such as ServerApplication.java, SecurityConfig.java, controllers, services, App.jsx, and components in VS Code.

1. Opening the main backend and frontend application source files in VS Code to view the full-stack code.

```

client > src > App.jsx > ...
37 import CertificatesHub from './components/CertificatesHub';
38 import ResumeBuilder from './components/ResumeBuilder';
39 import Internships from './components/Internships';
40 import Sessions from './components/Sessions';
41 import Forum from './components/Forum';
42 import Leaderboard from './components/Leaderboard';
43 import NotFound from './components/NotFound';
44 import InstructorDashboard from './components/InstructorDashboard';
45 import InstructorCourseStudents from './components/InstructorCourseStudents';
46 import InstructorCourseLessons from './components/InstructorCourseLessons';
47 import AdminDashboard from './components/AdminDashboard';
48 import ProtectedRoute from './components/ProtectedRoute';
49
50 function Landing() {
51   const [showSplash, setShowSplash] = useState(true);
52
53   return (
54     <>
55       {showSplash && <SplashIntro onComplete={() => setShowSplash(false)} />}
56       <Navbar />
57       <Hero />
58       <Routes />
59       <Instruments />
60       <Traillog />
61       <Testimonials />
62       <FAQ />
63       <Connect />
64     </>
65   );
66 }
  
```

1. Running the application in the terminal of VS code (./mvnw spring-boot:run and npm run dev).
2. Copying or clicking the local server link provided by Vite (<http://localhost:5173> / <http://localhost:8080>). The home page of the Enterprise Learning Platform — Enterprise Learning appears on the browser.

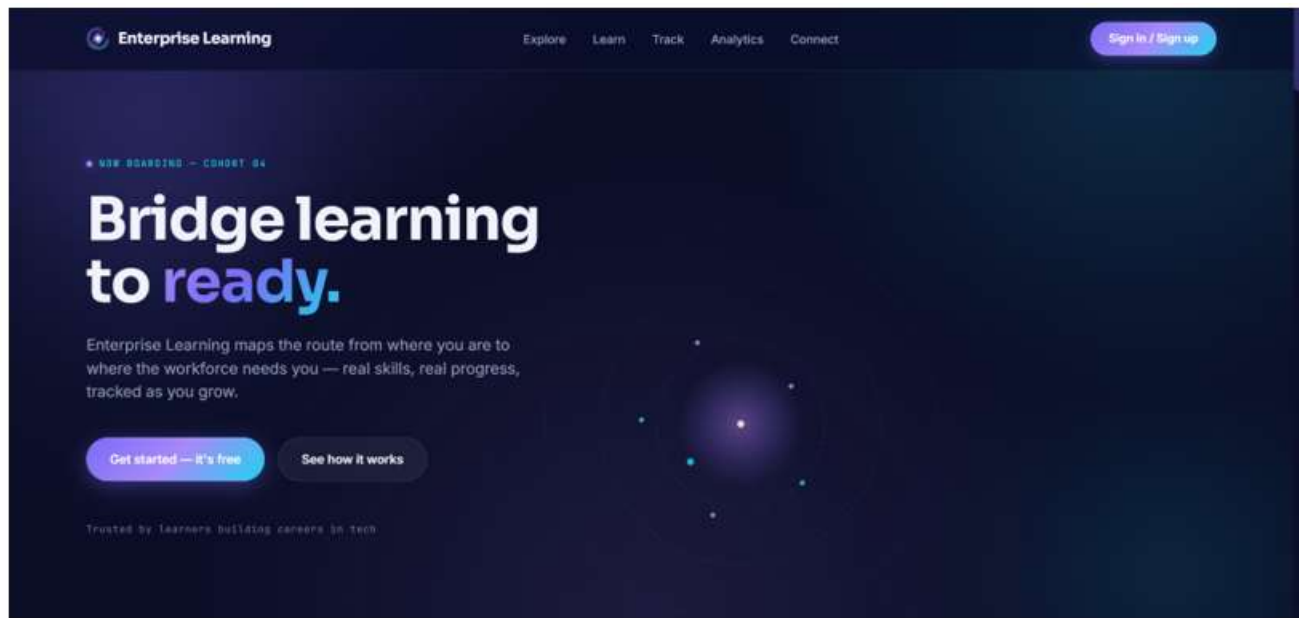
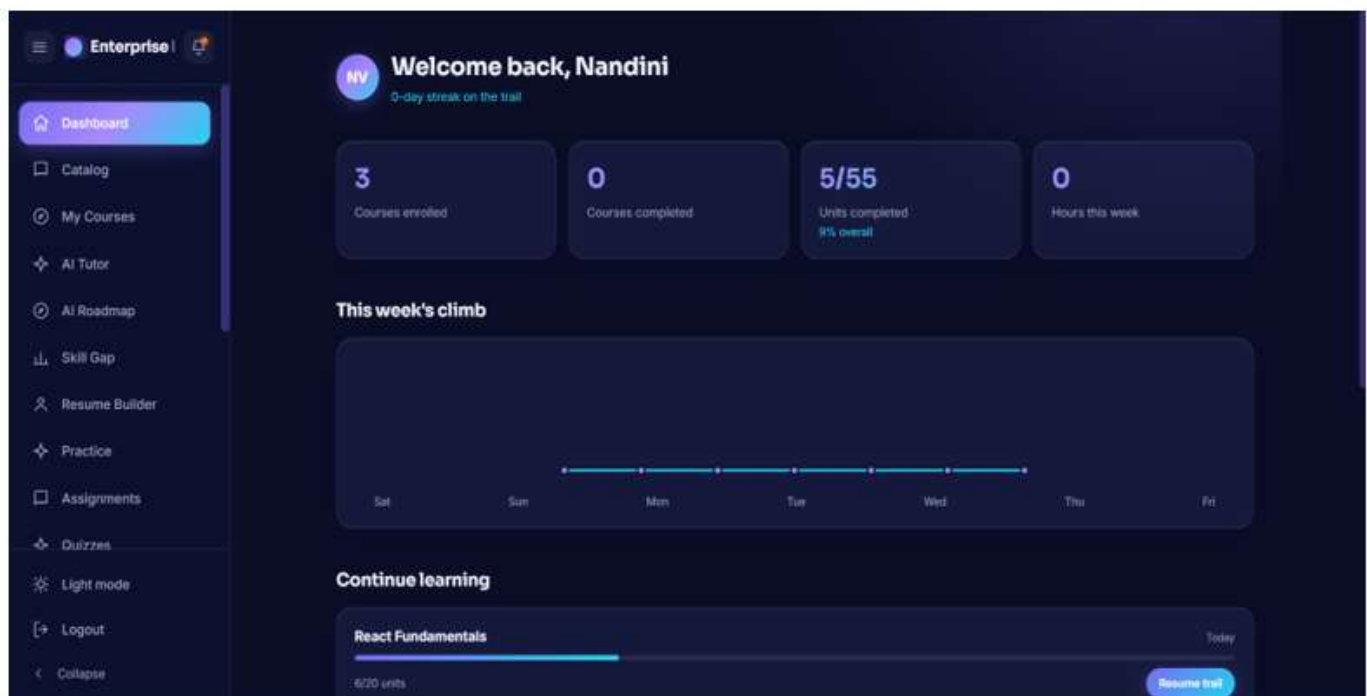


Figure 1: Landing Page Interface

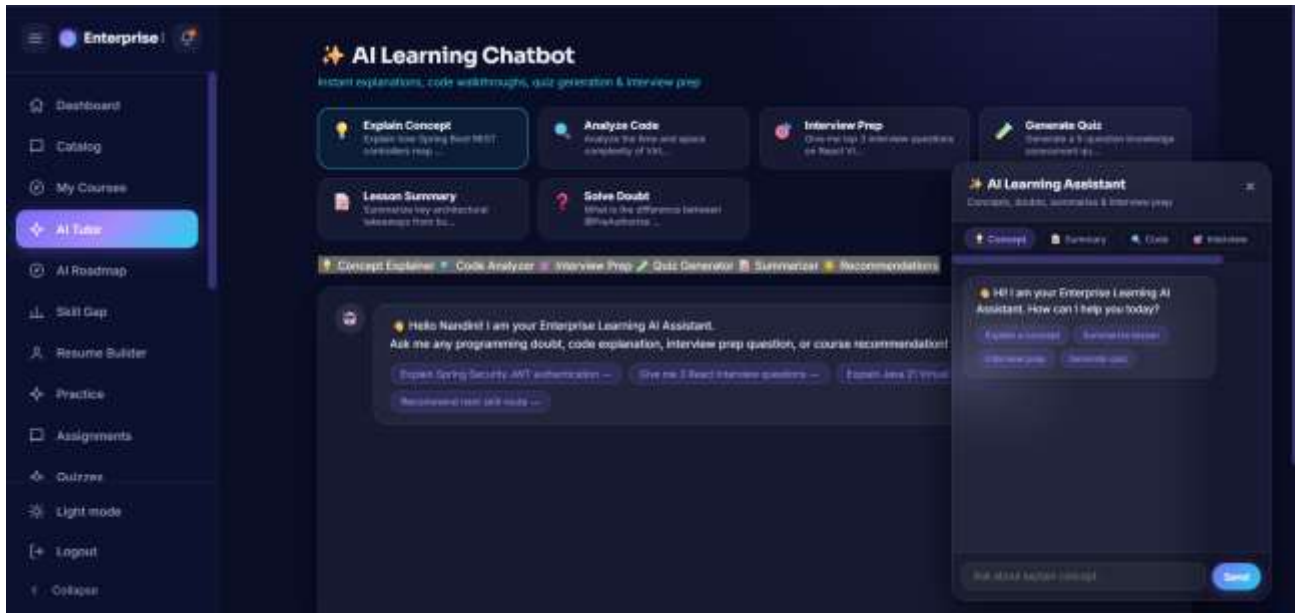
LANDING PAGE: Next page includes log in and also create an account for the new user or you can start your account with google also.

DASHBOARD PAGE:



AI CHATBOT & LIVE TUTOR PAGE:

Chatting with different AI tutors and code assistants is also available in the application.



7. Challenges Faced

During development of the wellness chatbot , several technical and operational challenges were encountered:

1. Integrating Machine Learning with the Web Application:

One of the main challenges was connecting the machine learning model with the Flask backend so that predictions could be generated dynamically based on user inputs.

2. Designing a User-Friendly Dashboard:

Creating an interactive and visually clear dashboard that displays wellness analytics in an understandable format required careful UI design and data visualization planning.

3. Handling Data Processing and Predictions:

Preparing the dataset and ensuring accurate predictions for mood, stress, and productivity using the machine learning model was a complex task.

4. Database Management:

Managing user data securely and maintaining proper database connectivity with SQLite while storing wellness records required careful implementation.

5. Chatbot Integration:

Integrating the chatbot system and ensuring it responds correctly to user queries related to wellness and mental health was another challenge.

6. Frontend and Backend Communication:

Ensuring smooth communication between the frontend interface and Flask backend through APIs required debugging and testing.

7. System Testing and Debugging:

During development, several issues such as server errors, data handling problems, and interface bugs were encountered and needed to be resolved through testing.

8. Ensuring Smooth User Experience:

Making sure that all modules such as login, dashboard, chatbot, and predictions worked seamlessly together was a significant challenge during the final integration phase.

8. Learnings & Skills Acquired

1. Java & Spring Boot Programming:

Understanding Java 17 and Spring Boot 3.5 was essential for developing the backend logic of the application and implementing REST services.

2. Spring Security & JWT Framework:

Knowledge of Spring Security helped in building the web application backend, managing routes, handling JWT authentication, and integrating OAuth2.

3. Frontend Development:

Skills in HTML, CSS, JavaScript, and React 19 were required to design responsive and user-friendly interfaces such as the landing page, login page, dashboard, and chatbot page.

4. Database Management:

Basic knowledge of SQLite was required to store and manage user information, wellness records, and system data efficiently.

5. Machine Learning Concepts:

Understanding machine learning concepts such as data preprocessing, model training, and prediction was necessary to build the wellness prediction model.

6. Scikit-learn Library:

The Scikit-learn library was used to develop and train machine learning models for predicting mood, stress levels, and productivity.

7. Data Visualization:

Knowledge of visualization tools like Chart.js helped in displaying wellness analytics through charts and graphs on the dashboard.

8. Problem Solving and Debugging:

Strong debugging and problem-solving skills were needed to fix errors, integrate different components, and ensure smooth functionality of the system.

9. Testimonials from team**Team Member 1:(M.Madhumitha)**

The project helped in improving analytical thinking and practical knowledge in web development, machine learning integration, and data visualization while working on a real-world wellness application.”

Team Member 2:

“Working on this project strengthened problem-solving abilities through tasks such as backend development, data processing, and implementing machine learning models for wellness prediction.”

Team Member 3:

“The development of the Learning platform enhanced technical skills in Java Flask, database management, and AI-based system design while providing experience in building real-world applications.”

10. Conclusion

The Enterprise Learning Platform — Enterprise Learning project successfully demonstrates the integration of web development and artificial intelligence to create an intelligent learning monitoring system. The platform allows users to track learning-related data such as course progress, quiz scores, and study streaks, while providing predictions and insights through an AI model. The system also includes an interactive dashboard, data visualization, and an AI chatbot to support users in understanding their learning patterns. Through this project, practical knowledge of React 19, Spring Boot, MySQL database management, and machine learning/AI was applied to build a real-world application. Overall,

Acknowledgements

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